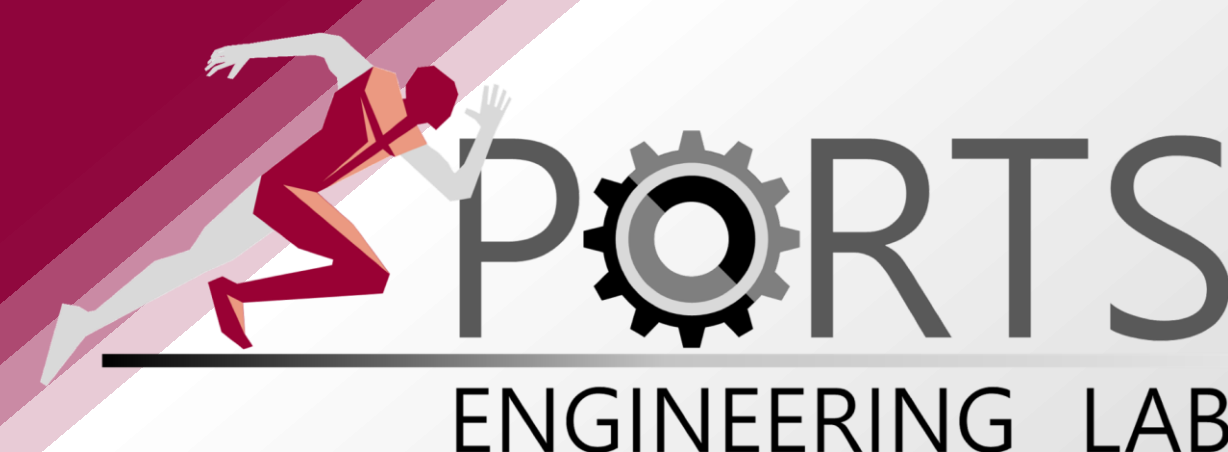




CHIHYEONG LEE
chi0412@snu.ac.kr
Advisor: Dr. Joeun Ahn

Looking for:
POSTDOC. POSITION

Evaluating, Understanding, and Predicting Human Movement



EDUCATION

Seoul National University, Sports Engineering Lab

- Ph.D. in Sports Science (*Expected Feb 2027*)
- M.S. in Sports Science, 2023

University of Houston, Health and Human Performance

- Visiting Scholar, 2024

SKILLS

Experimental Systems

- Motion Capture Systems (Cameras & IMUs), Wireless EMG, Instrumented Split-Belt Treadmill, Force Plates

Modeling & Simulation

- MATLAB, Python
- Visual3D, OpenSim, SCONE, Hyfydy

GOVERNMENT-FUNDED RESEARCH

- **Ministry of Trade Industry and Energy (MOTIE)**
 - Industrial wearable robots (~USD 2M, 2021–2023)
 - Healthcare wearable robots (~USD 6M, 2022–2025)
- **Ministry of Health & Welfare (MOHW)**
 - Wearable robot evaluation (~USD 1M, 2023–2027)
- **Ministry of Science and ICT (MSIT)**
 - Fall Prevention (~USD 110K, 2022–2023)
 - Stroke Gait Rehabilitation (~USD 280K, 2024–2026)



The Government of
the Republic of Korea

“Contributed to successful grant proposals, led technical development and delivered project outcomes.”

PUBLICATIONS

- **JOURNALS:** 4 first-author journals (1 under review)
- **CONFERENCES:** 3 peer-reviewed papers, 6 abstracts

CONTACT



Curriculum
Vitae



GitHub Pages



SPORTS
ENGINEERING LAB

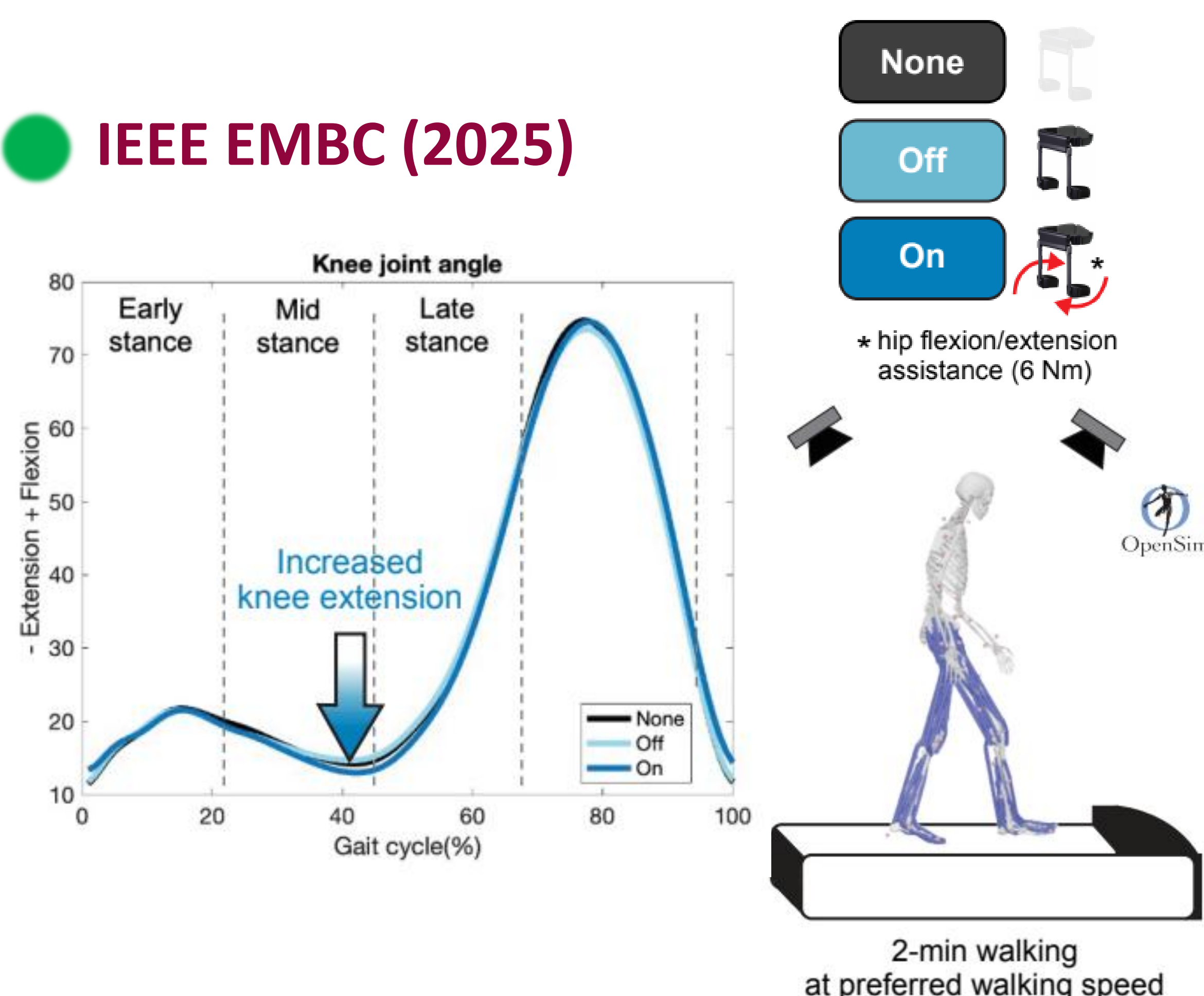
HUMAN-ROBOT INTERACTION

- **IEEE RA-L (2025)**
IEEE Robotics and Automation Letters (IF 5.3)



*“The proposed fully **soft lift-assist wearable suit** showed the potential to **reduce muscle activation and metabolic cost during lifting tasks.**”*

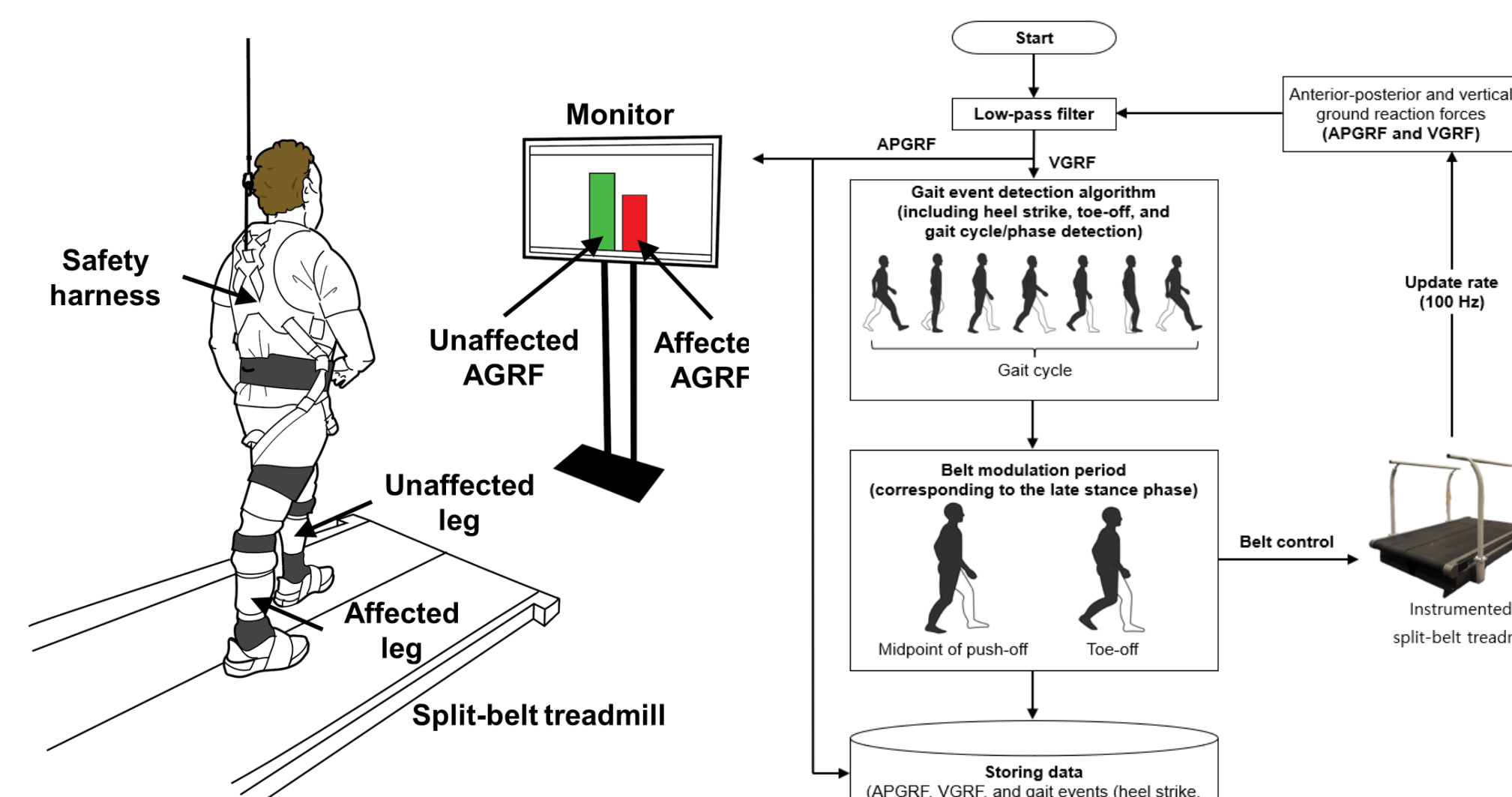
- **IEEE EMBC (2025)**



*“Vector coding technique revealed that a **hip assist robot altered hip-knee coordination**, particularly during the mid-to-late stance phase.”*

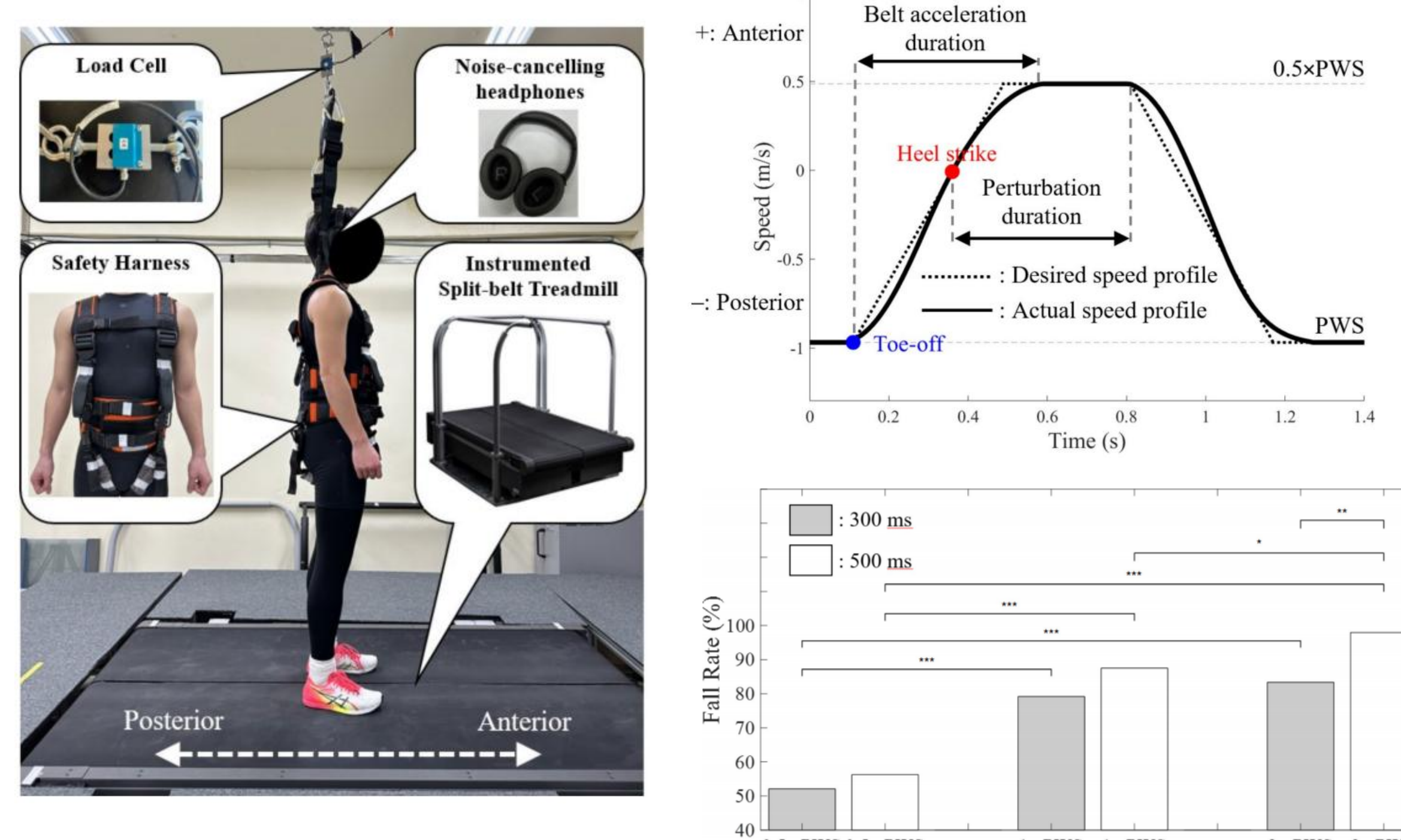
HUMAN RESPONSE ANALYSIS

- **JNER (2026, under review)**
Journal of NeuroEngineering and Rehabilitation (IF 6.0)



*“We developed a **real-time post-stroke gait training system by customizing an instrumented split-belt treadmill. Visual biofeedback and adaptive belt deceleration synergistically enhanced gait propulsion.**”*

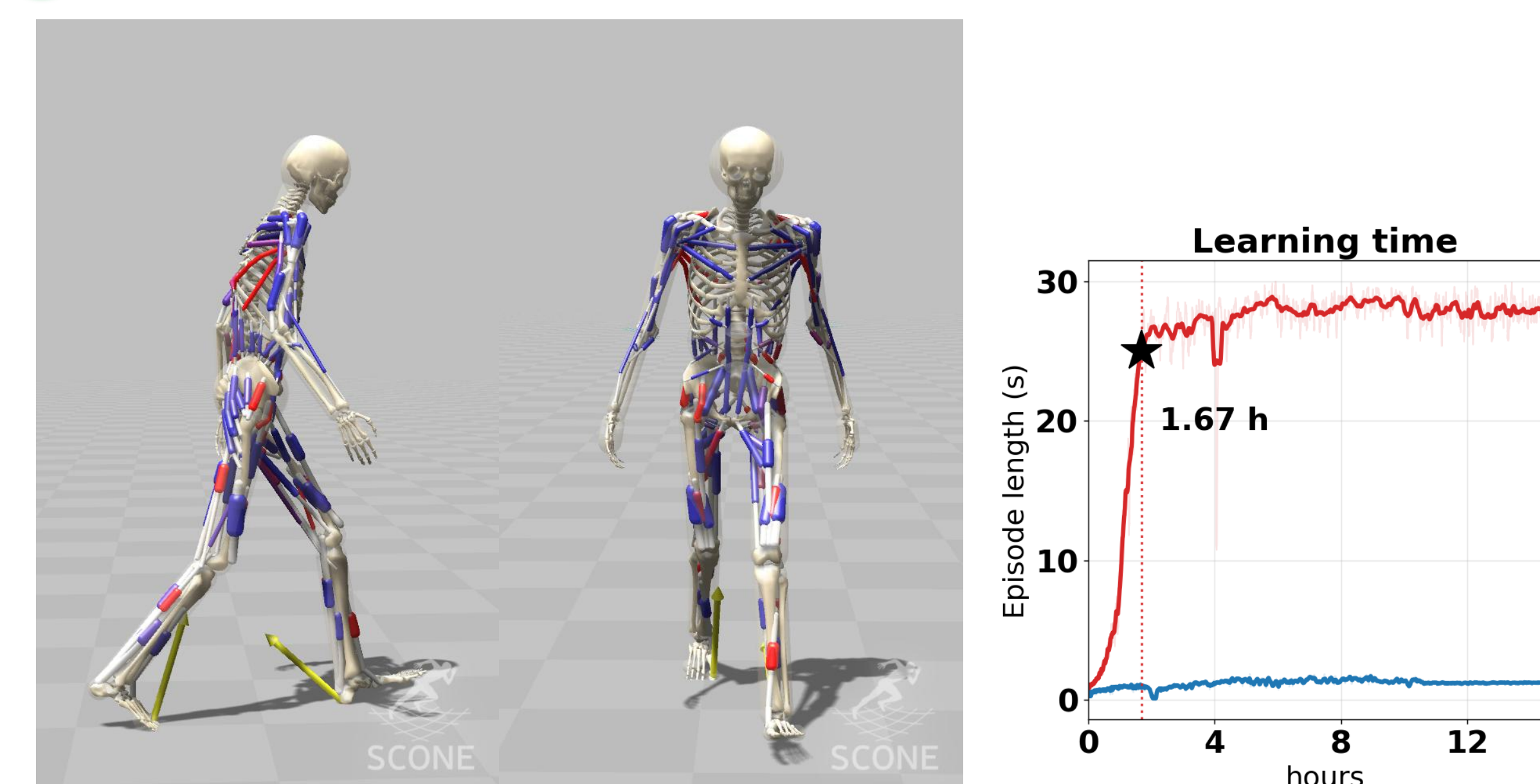
- **Scientific Reports (2025) (IF 4.9)**



*“We customized an instrumented split-belt treadmill to deliver **real-time slip perturbations** and establish **design criteria for fall-inducing experiments.**”*

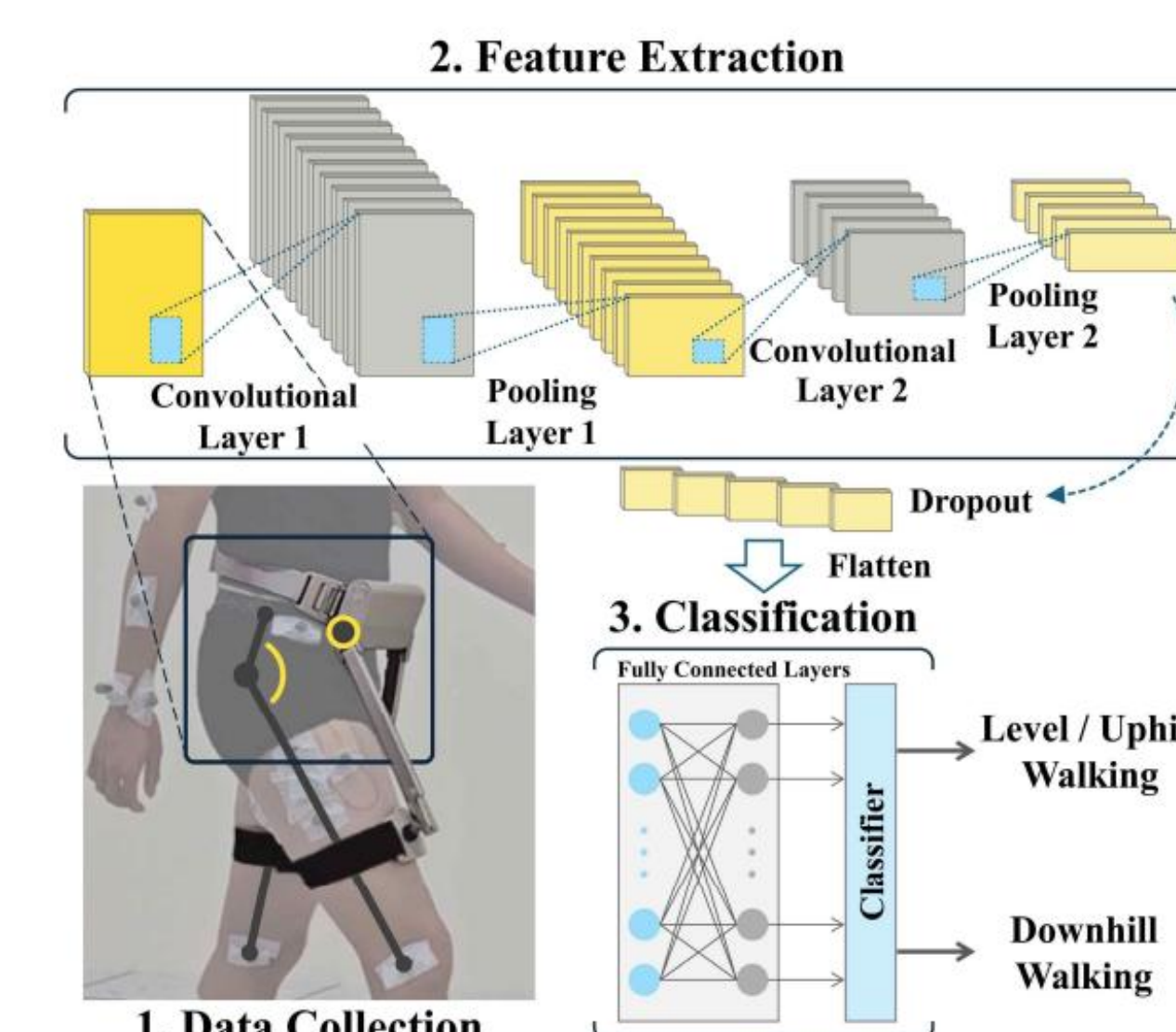
HUMAN MOVEMENT PREDICTION

- **WCB (2026)**



“Reference Configuration Imitation Learning (RCIL) achieved fast and stable walking by learning residual posture corrections rather than direct muscle activations.”

- **ASB (2025)**



*“A **deep learning** approach enabled **accurate and real-time classification of downhill walking** for adaptive gait-assistive wearable robots.”*

- **ISBS (2023)**

*“Machine learning enabled **accurate and real-time detection of backward slips** for safer gait-assistive wearable robots.”*